

Determining critical phase transition temperature in an Ising model simulation

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TODO: <Abstract>

I. INTRODUCTION

A. The Ising model

The Ising model named after the physicist Erns Ising is a model in statistical mechanics. The model consists of discrete magnetic spin dipole moments that take values ± 1 ("up" and "down" respectively) arranged on a lattice. For the purposes of our simulation we used a square lattice. The spins interact with only a subset of all the spins that are "close", we call this subset the neighbouring spins. It is known that this model exhibits phase transitions that can be associated with the gas-liquid transition or spontaneous magnetisation. [1]

B. Aim of study

In this article we look at determining at what temperature does the Ising model undergo a phase transition. It is known from theory that below a certain critical temperature and after sufficient amounts of time, the model stabilises into completely up or down state. Conversely, above this critical, on average, the model remains a random mixture of up and down spins. [1] We will try to determine this value by simulating the Ising model for different temperatures, differentiate between ordered and random results, and determine the critical value at which the phase transition happens.

II. THEORY

In the Ising model, the lattice spin elements take the values $s_i = \pm 1$. The interaction energy between two spins is given by

$$E_{\text{int}} = -J s_i s_j, \quad (1)$$

where $J > 0$ is the coupling constant of ferromagnetic interactions and the s_i being the value of spin i . We define the reduced interaction energy as $E_r = \frac{E_{\text{int}}}{J} = -s_i s_j$. The reduced interaction energy with the j nearest neighbours is then

$$E_r = \sum_{\langle j \rangle} s_i s_j, \quad (2)$$

The energy difference between the current spin at some position and the energy after the spin has been flipped is

given by

$$\Delta E = E_{\text{before}} - E_{\text{after}} = \sum_{\langle j \rangle} (-s_i) s_j - \sum_{\langle j \rangle} s_i s_j = -2 \sum_{\langle j \rangle} s_i s_j. \quad (3)$$

The total amount of energy of the whole spin lattice is given by the sum over all pairwise interaction energies. This is then denoted as

$$E_{\text{sys}} = -J \sum_{\langle i, j \rangle} s_i s_j, \quad (4)$$

where the sum over i, j counts every pair only once with $(i, j) \equiv (j, i)$. The reduced system energy is then given by

$$E_{\text{sys-red}} = \frac{E_{\text{sys}}}{J} = - \sum_{\langle i, j \rangle} s_i s_j, \quad (5)$$

We then define another dimensionless quantity, the reduced temperature as

$$T_r = \frac{k_B T}{J}, \quad (6)$$

where T is the physical temperature, k_B is the Boltzmann's constant and J the coupling constant. We then only work in terms of this reduced temperature and give no physical values of T or J .

The theoretical value of the critical reduced temperature was predicted by Lars Onsager to be

$$T_c = \frac{1}{H_c} = \frac{1}{\frac{1}{2} \ln(\cot(\frac{\pi}{8}))} = \frac{2}{\ln(1 + \sqrt{2})}, \quad (7)$$

where $H = J_r$ is the reduced coupling constant and H_c is the critical coupling constant[2].

Later, in 1952, Yang Chen-Ning showed that the function of absolute value of average magnetisation as a function of reduced temperature is given by

$$|\langle m \rangle|(T_r) = \left[\frac{1 + x^2}{(1 - x^2)^2} (1 - 6x^2 + x^4)^{\frac{1}{2}} \right]^{\frac{1}{4}}, \quad (8)$$

with $x = e^{-\frac{2}{T_r}}$ [3].

The heat capacity of the Ising model is given by

$$C = k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2), \quad (9)$$

where $\beta = \frac{1}{k_B T}$ [1]. We can modify to equation to get an expression for the heat capacity in terms of the reduced

energy E_r and reduced temperature T_r as

$$\begin{aligned} C &= k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2), \\ &= k_B \frac{1}{k_B^2 T^2} (\langle E^2 \rangle - \langle E \rangle^2), \\ &= k_B \frac{1}{J^2 T_r^2} (\langle E^2 \rangle - \langle E \rangle^2), \\ &= k_B \frac{1}{J^2 T_r^2} \mathcal{J}^2 (\langle E_r^2 \rangle - \langle E_r \rangle^2), \\ &= k_B \frac{(\langle E_r^2 \rangle - \langle E_r \rangle^2)}{T_r^2}. \end{aligned}$$

We define the dimensionless reduced heat capacity as

$$C_r = \frac{C}{k_B} = \frac{(\langle E_r^2 \rangle - \langle E_r \rangle^2)}{T_r^2}. \quad (10)$$

We know that the heat capacity has a singularity at the critical temperature where it becomes infinite[2].

An approximation of the heat capacity around the critical temperature is given by

$$C \approx B - A \ln \left(\sqrt{(T - T_c)^2} \right). \quad (11)$$

III. METHODOLOGY

A. Metropolis algorithm

To simulate the evolution of the Ising model we use a Monte Carlo simulation using the Metropolis algorithm. In our implementation we consider the von Neumann neighbourhood of only 4 neighbouring spins in the $\pm x$ and $\pm y$ directions. We also use a periodic boundary condition such that if we denote by $n \times n$ the size of the lattice, we have that the spin at position (n, i) is a neighbor of spin at position $(1, i)$ and in the same way also spin at i, n is neighbouring the spin at $(i, 1)$. At each iteration of the algorithm a random position is chosen in the square lattice of spins. The energy needed to flip the spin from the up to the down state or vice versa is then calculated from Eq. (3). We then have 2 cases. If $\Delta E \leq 0$, we flip the spin orientation. If $\Delta E \geq 0$, we accept the flip with probability $p = e^{\frac{\Delta E}{T_r}}$. In order to do this, we sample a random variable $X \sim \text{Unif}(0, 1)$ and check $X < p$, in which case we accept the flip and leave the spin unflipped otherwise.

Theory tells us that for values of the reduced temperature below the critical temperature i.e. $T_r < T_c$ after sufficient spin-flip attempts, the system gets to an equilibrium state where all the spins are either up or down and therefore the absolute value of the average magnetisation goes to 1[1]. To calculate the average quantities $|\langle m \rangle|$ and C_r we then wait for a certain amount of spin-flip attempts before starting to calculate them, so that we measure the equilibrium of the model.

B. Determining the Critical Temperature

To determine the critical temperature, there are a few options. As given in the theory, the two major events happening at this temperature is the switch of the absolute value of the average magnetization from 1 to 0, and the spike in the heat capacity. Since we are still working with some randomness, we decided that it would technically be easier to find the peak of the heat capacity. In our simulation setup, we are not finding a lot of temperatures, but we are running a lot of simulations for each temperature. This means that we do not want to use a peak finding algorithm, but rather fit a function to the data and find the maximum of that function. To determine the critical temperature, we use a fit function over the heat capacity. We use a modified version of Eq. (11), where we remove the singularity and end up with a finite peak. The formula is given by

$$C \approx B - A \ln \left(\sqrt{(T - T_c)^2 + \omega^2} \right). \quad (12)$$

This is implemented in the code as a function that takes the parameters A, B, T_c, w and the temperature T_r and returns the value of the heat capacity at that temperature. We then use the scipy package to fit this function to the data we have collected. The parameters A, B, T_c, w are then optimized to minimize the difference between the function and the data. The parameter T_c is then the critical temperature we are looking for. The peak of this function then gives us the critical temperature, which we can then compare to the theoretical value given in Eq. (7).

C. Determining parameters for final simulation

For the simulation that produced Figure 1 we had a number of requirements. The first requirements was for the simulation that produces the figure we use in this article to take no longer than 20 minutes to run. This meant that we had to look for simulation parameters that produce good data but for the simulation runtime to stay within the time bound.

We already knew from test simulations that the critical temperature was somewhere in the range $1 < T_c < 3$ as we ran simulation for $T_r = 1.5$ and $T_r = 3$ and observed the behaviour associated with reduced temperatures being lower and higher than the critical temperature respectively. This then set the range of reduced temperatures that we were going to simulate.

We then continued to choosing a size of the lattice that we want to simulate as this, we found, had impact on the number of spin-flip attempts needed for the model to reach equilibrium. We decided to use $n = 50$ as for values of $n < 20$ we found that the equilibrium was disturbed to a great degree by the non-zero value of the temperature causing fluctuations in the spins.

We then ran multiple simulations trying to ascertain the

number of spin-flip attempts needed for the 50×50 lattice to reach equilibrium for the temperature range we wanted to study. We found that the configuration that took the most amounts of spin-flip attempts to reach equilibrium was the one with the lowest temperature, namely $T_r = 1.5$. We decided on the number $k = 27'000'000$ for the amount of spin-flip attempts to wait before collecting values of the system energy and magnetisation for the average magnetisation and heat capacity calculations.

We then collect 500 of these quantities to average, one every 27'000 spin-flip attempts.

Finally, we tried to run as many individual simulations for every temperature value we were interested in, while also doing the simulations for a decent enough number of different temperature values to get a good resolution in our final results. We decided to do 42 simulations for each of 43 different temperature values.

D. Uncertainties

The uncertainties in our results are calculated using the standard deviation of the mean. We calculate the standard deviation of the values we collected for the average magnetisation and heat capacity over the 49 simulations per temperature value. This gives us an estimate of the uncertainty in our results.

We also have to consider the uncertainty in our fitting function. We use the scipy package to fit our function to the data, and it provides us with an estimate of the uncertainty in the fitted parameters. We can then use this to estimate the uncertainty in our critical temperature value that is working with the uncertainties of the fit.

IV. RESULTS

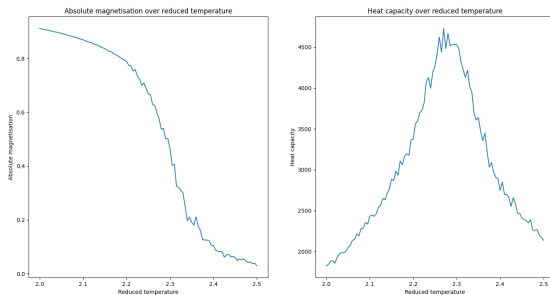


FIG. 1. Results

TODO: <Give results>

TODO: <Give figures of results>

V. DISCUSSION

We were able to determine the critical temperature of the Ising model using a Monte Carlo simulation with the Metropolis algorithm. Our results show that the critical temperature is approximately $T_r = \langle \text{TODO} \rangle$, which is in good agreement with the theoretical value of $T_c = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$. The graphical representation of the results clearly illustrates the behavior of the system as it approaches the critical temperature, with the heat capacity diverging and the average magnetization dropping to zero. Nevertheless, the high uncertainties in combination with the low reduced chi-squared value of the fit indicate that we overestimated the uncertainties. This could be due to a number of factors, including the limited number of simulations and the choice of fitting function.

The difference between theoretical and experimental values of the critical temperature as well as the wrongly calculated uncertainties might however not be due to a simulation error. We can see in the graphical representation that the heat capacity peak is not as sharp as in the theoretical prediction. This can most likely be attributed to the finite size of the lattice used in the simulation. In a finite system, the phase transition is not as sharp as in an infinite system, leading to a rounding of the heat capacity peak and a shift in the critical temperature. This is also being discussed by M.Yu. Malsagov, I.M. Karandashev and B.V. Kryzhanovskiy[4], who propose to add a correction term to the heat capacity formula, which reduces the systematic error of the finite lattice approximation. The result of this inaccuracy is that the critical temperature we determined from our simulation is slightly lower than the theoretical value as well as that the uncertainties are underestimated. This is consistent with the expectation that finite-size effects lead to a shift in the critical temperature.

TODO: <Compare our results with theoretical values>

VI. CONCLUSIONS

While we were able to determine a value for the critical temperature, we see that our results do not correspond to the theoretical value by 100%. Further research could explore different fitting functions or increase the number of simulations to improve the accuracy of the results. Additionally, exploring different lattice sizes or boundary conditions could provide further insights into the behavior of the Ising model and its phase transitions. Another option would be combining different approaches for the critical temperature, such as looking at the average magnetisation or using a peak finding algorithm. This would then result in a more robust value for the critical temperature.

There is a way to improve the accuracy of the finite lattice approximation that we are using with a correction term for the heat capacity formula as proposed by M.Yu. Malsagov, I.M. Karandashev and B.V. Kryzhanovsky[4]. They show that with their improvement, the difference between experimental and theoretical value for the heat capacity using a 50×50 matrix is less than the statisti-

cal error of the monte carlo simulation itself. This would mean that the systematic error of the finite approximation is irrelevant compared to the inherent error of the simulation method itself. Overall, our findings validate the theoretical predictions and demonstrate the effectiveness of Monte Carlo simulations in studying phase transitions in statistical mechanics.

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